

Erdős Problem #30

EP-30

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $h(N)$ be the maximum size of a Sidon set in $\{1, \dots, N\}$. Is it true that, for every $\epsilon > 0$, $h(N) = N^{1/2} + O_\epsilon(N^\epsilon)$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>